Analysis of Various Aspects of Music

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Introduction

Most people are interested in the music industry. Recent studies and surveys have revealed that 86 to 90 percent of people say they love listening to music. Research shows that music can relieve stress and improve people's mood. Music has become a significant influence in people's daily lives. One's ability to focus is aided by music, and increased productivity raises the calibre of output and raises standards. As it gained popularity, platforms offering a diversity of music started to appear, steadily increasing the number of listeners.The development of music platforms also provides data on a variety of musical qualities, including vitality, volume, key, and liveliness. This examination looks at how various musical qualities relate to one another. In this examination, the interaction between distinct musical parts and how they improve a song's vibrancy is the main focus.

Objectives

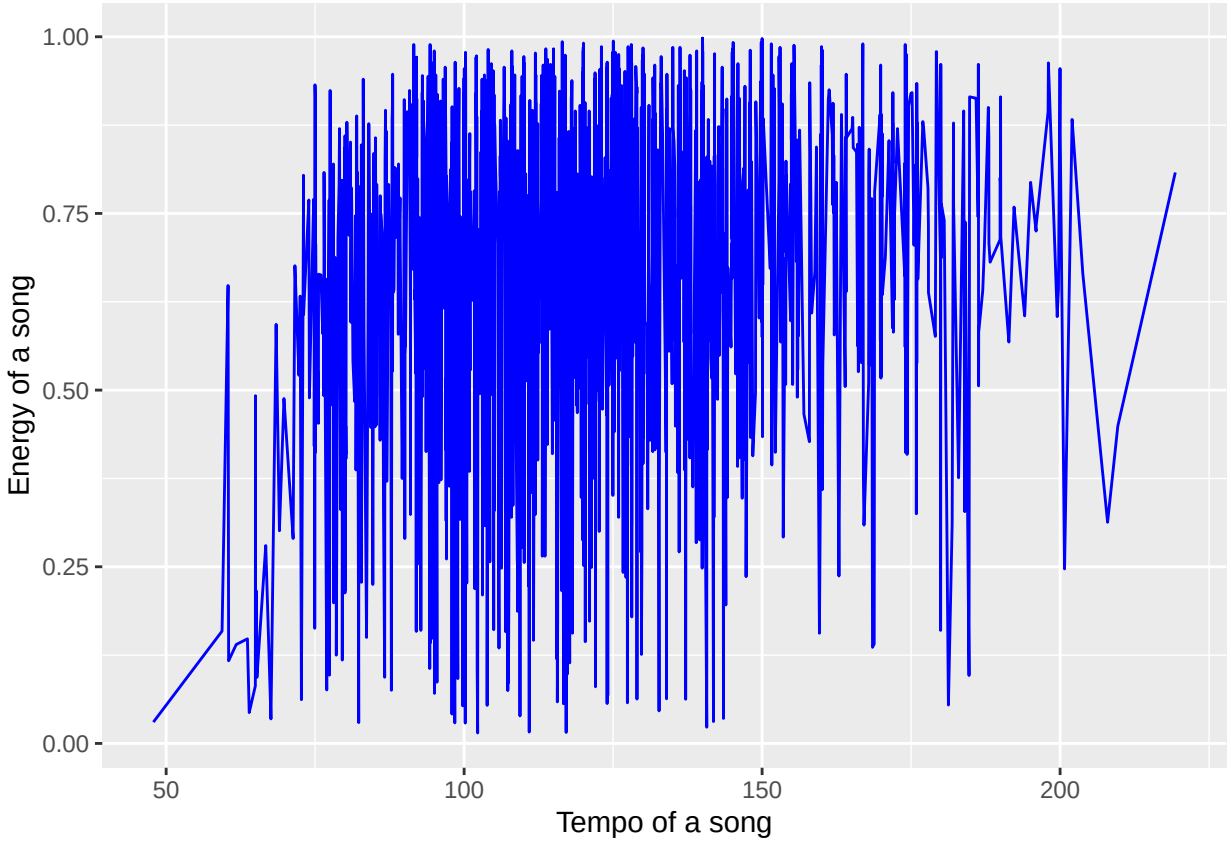
creating a line plot to comprehend the relationship between "tempo" and "energy," as well as the relationship between "loudness" and "energy." Using hierarchical modelling, examine the relationship between the variables "danceability" and "liveness." Finding the relationship between the variables "liveness" and "danceability" using Hypothesis.

Data

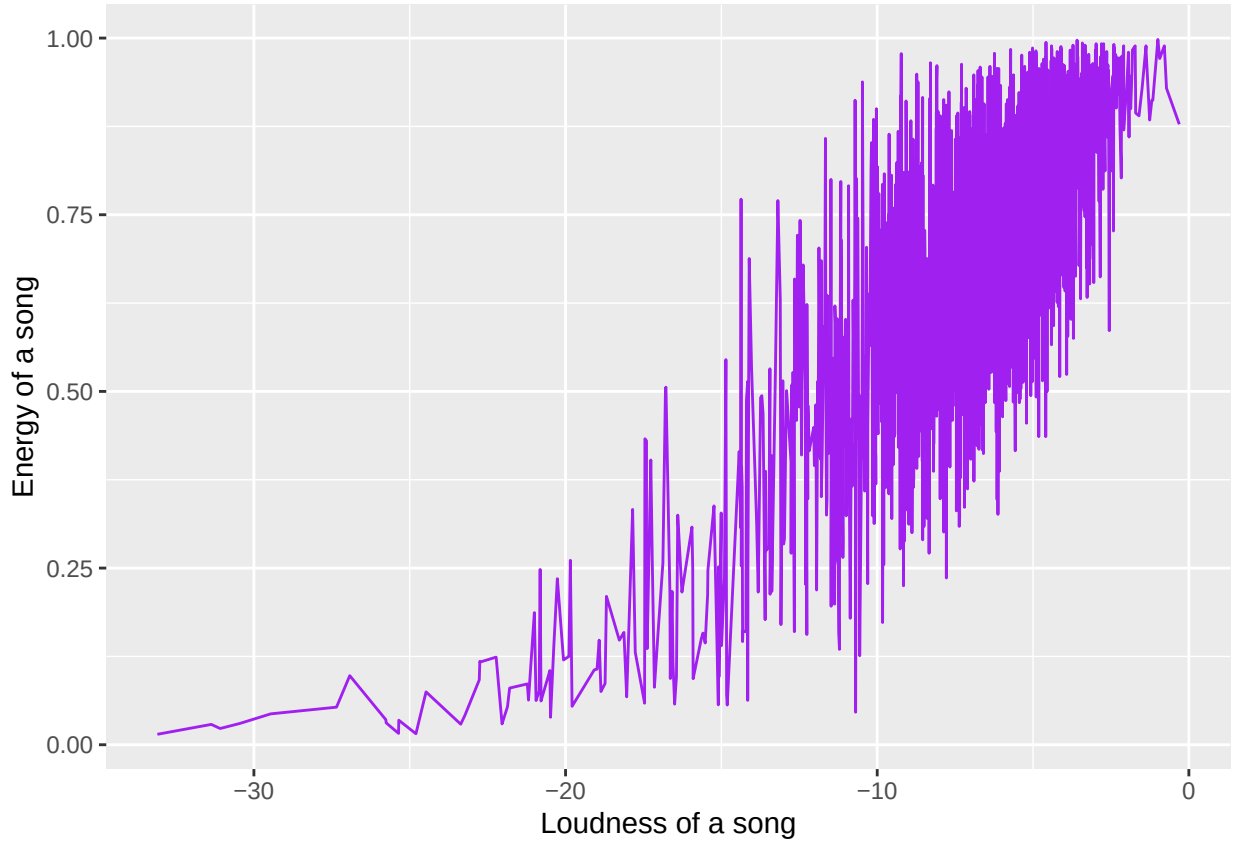
Spotify music analysis was utilised to acquire the data for this investigation. The material provides details about 2017 songs in several categories, such as "liveness," "energy," "key," "loudness," "danceability," etc. The analytical data was obtained from the music streaming service "Spotify," and the data set was compiled from https://www.kaggle.com/code/aeryan/spotify-music-analysis

Analysis and Results

Analysis of relation between tempo and energy of a song:-



Analysis of realation between loudness and energy of a song:-



The line plots above demonstrate that tempo is not always proportional to the song's vitality. This is clear from the line graphs. A slower tempo does not necessarily translate into a song with lower vitality. Although a song with a tempo of 50 or less can always produce low energy music, the amount of energy that can be reduced by a song with a very high tempo is constrained. despite the fact that tempo and energy are not directly connected.

However, there is a consistent relationship between volume and energy. The line plot above makes it clear that a song's energy steadily grows as loudness increases. This leads us to the conclusion that a song's volume and energy are generally inversely correlated. Energy rises in direct proportion to loudness.

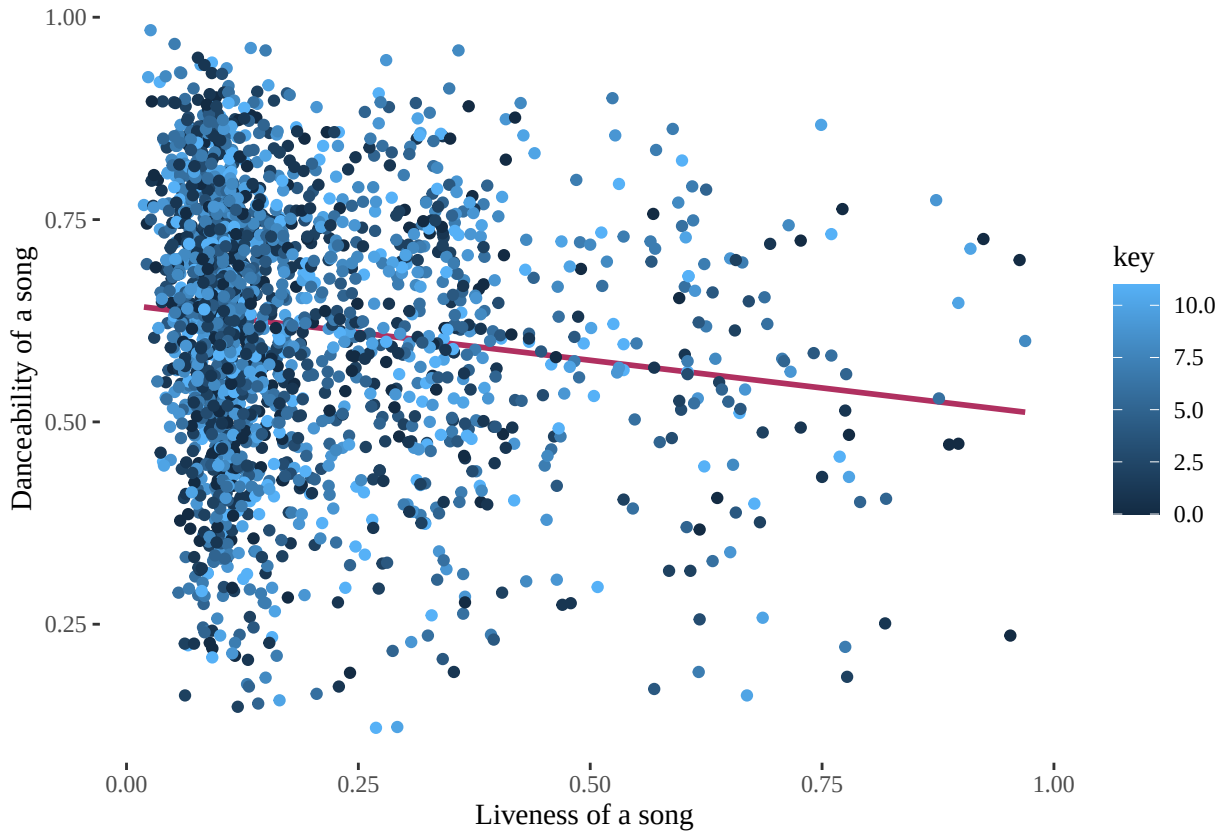
Hierarchical modelling:-

This is done with variables 'liveness' and 'danceability'

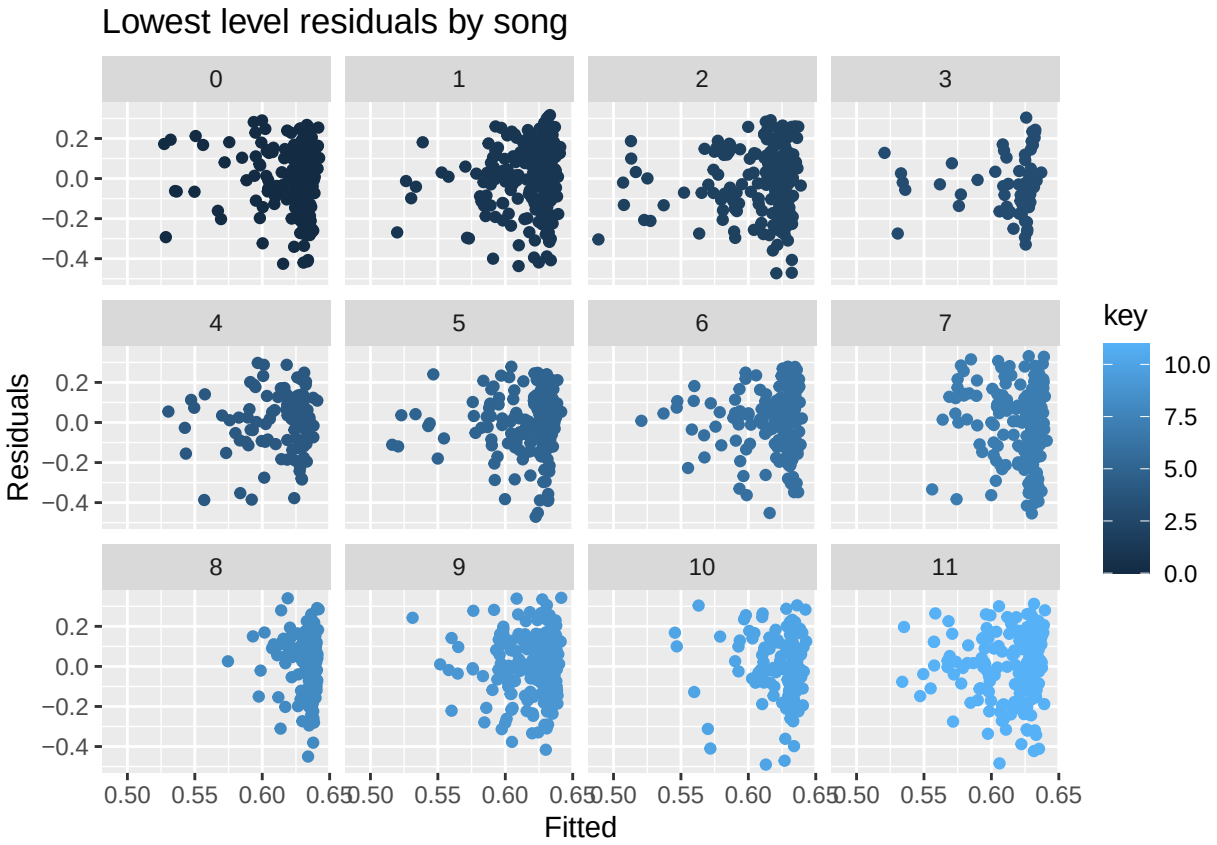
```
##      Aco danceability    Dur energy key liveness loudness speechiness   tempo
## 1 0.0102        0.833 204600  0.434   2   0.1650   -8.795      0.4310 150.062
## 2 0.1990        0.743 326933  0.359   1   0.1370  -10.401      0.0794 160.083
## 3 0.0344        0.838 185707  0.412   2   0.1590   -7.148      0.2890  75.044
## 4 0.6040        0.494 199413  0.338   5   0.0922  -15.236      0.0261  86.468
##   valence           song           artist
## 1   0.286       Mask Off           Future
## 2   0.588        Redbone Childish Gambino
## 3   0.173   Xanny Family           Future
```

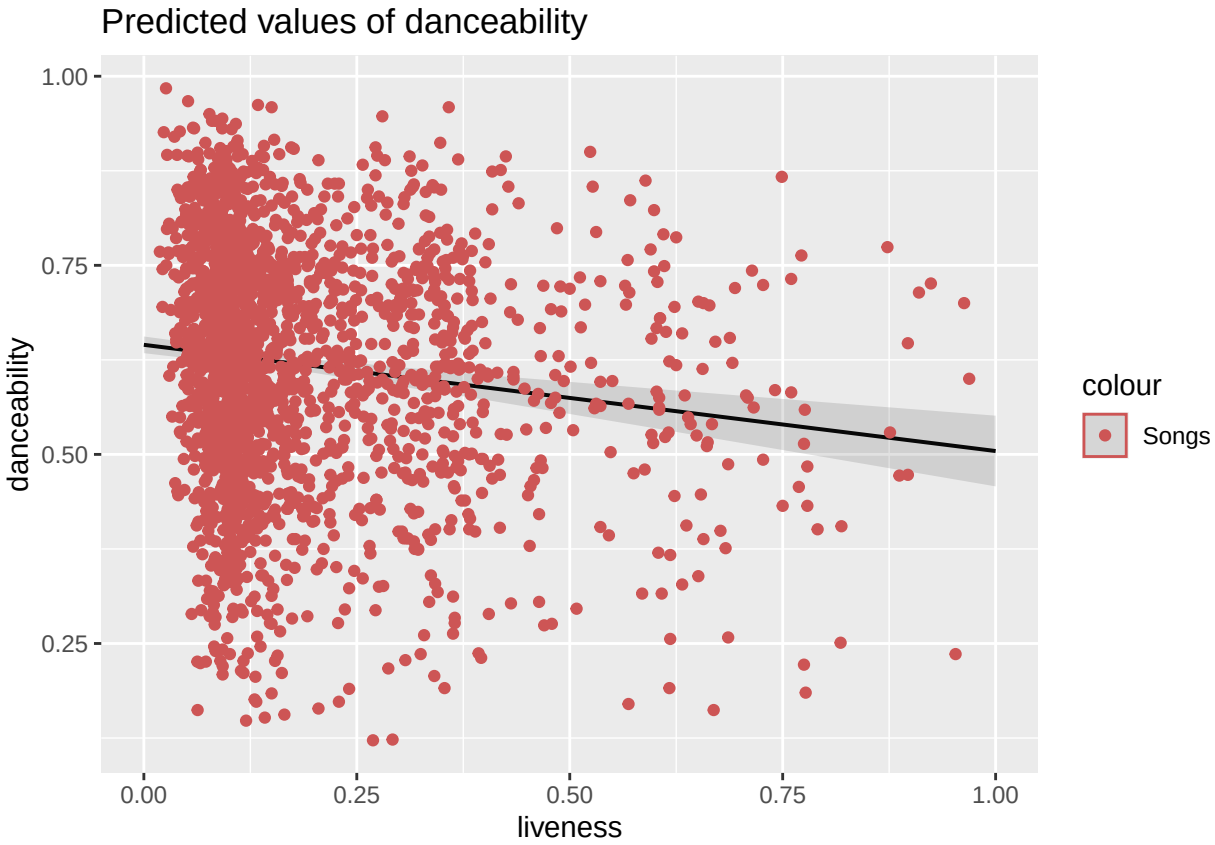
```
## 4   0.230 Master Of None      Beach House
```

Plotting:-



From the plot it can seen that most of the songs that have low 'liveness' have high 'danceability' and as the liveness of a song increases the danceability decreases.


Lowest level residuals by song
0
1
2
3
4
5
6
7
8
9
10
11
Residuals
0.2
0.0
−0.2
−0.4
0.50
0.55
0.60
0.65
Fitted
key
10.0
7.5
5.0
2.5
0.0



According to the study generated from hierarchical modelling, a song's danceability(a combination of song qualities including beat intensity, tempo stability, and overall pace)decreases as the liveness(the presence of an audience in a song)increases. Most songs with low liveness have high rates of danceability. A conclusion drawn from the investigation is that liveliness and danceability are adversely related. When a survey was done to gauge danceability as liveliness increased, the results showed a drop.

Hypothesis testing

H0:Liveness and Danceability are propotional H1:Liveness and Danceability are inversely propotional

```
##
##  Shapiro-Wilk normality test
##
## data:  newdata_2$danceability
## W = 0.98475, p-value = 7.565e-14
```

```
##
##  Shapiro-Wilk normality test
##
## data:  newdata_2$liveness
## W = 0.77597, p-value < 2.2e-16
```

Table 1: Pearson's Table

estimate	statistic	p.value	conf.low	conf.high	method	alternative
-0.13	-5.98	0	-0.17	-0.09	Pearson's product-moment correlation	two.sided

According to the hypothesis, "liveness" and "danceability" do not have a normal distribution, which prompted an investigation using the Pearson's correlation coefficient. From the computation, it can be shown that the variables "liveness" and "danceability" have a very modest relationship and a negative correlation, which supports the alternative hypothesis that they are inversely proportional.

Limitations

In the dataset, the songs from a certain time period is included. The results are contrained to a certain time period and it might vary for songs from different time. The dataset not only varies with time but also varies from people to people as the preferences of songs changes. Thus the analysis will produce conflicting results for diverse people.

Conclusion

The line plot demonstrates that the song's speed is not always proportional to its intensity. The line plot also demonstrates the strong likelihood of a relationship between loudness and energy being proportionate. The link between liveness and danceability is inversely proportional, according to the hierarchical modelling. It is also acknowledged that the presented plots' and hierarchical modeling's conclusions may alter as a result of future data. Overall, there is a correlation between the factors "loudness and energy" and "likeability and danceability." The results of the experiment show that the variables "liveness" and "danceability" have a very weak relationship and a negative association.

APPENDIX

Packages Used

```
require(lme4)
require(lmerTest)
require(tidyverse)
require(readxl)
require(magrittr)
require(sjPlot)
require(sjmisc)
require(sjstats)
require(arm)
require(amelia)
require(pdftools)
require(ggplot2)
```

Importing required data for analysis

```
mydata <- read.csv("C:/Users/Elena Vallikappan/Downloads/data (1).csv")
View(mydata)
```

#Excluding columns that are not required

```
myvars <- names(mydata) %in% c("X","instrumentalness","mode","time_signature","target")
newdata <- mydata[!myvars]
```

#Analysis of relation between tempo and energy of a song:-

```
# constructing line plots
# Line plot of music using ggplot2
ggplot(data=newdata, aes(x=tempo,y=energy)) + # ggplot with the desired data
geom_line(colour="blue") + # Specifying a bar chart
labs(x="Tempo of a song",y="Energy of a song") # Axes labels
```

#Analysis of realation between loudness and energy of a song:-

```
# constructing line plots
# Line plot of music using ggplot2
ggplot(data=newdata, aes(x=loudness,y=energy)) + # ggplot with the desired data
geom_line(colour="purple") + # Specifying a bar chart
labs(x="Loudness of a song",y="Energy of a song") # Axes labels
```

Hierarchical modelling:-

#This is done with variables ‘liveness’ and ‘danceability’

```
head(newdata)
 #wrangling
newdata_2 <- rename(newdata,Aco="acousticness",
Dur = "duration_ms",
song = "song_title")
head(newdata_2[1:4,])
datalmer1 <- lmer(danceability~liveness + (1|key) ,data=newdata_2)
summary(datalmer1)
```

Calculating ICC

```
datalmer1vc <- VarCorr(datalmer1)
print(datalmer1vc,comp="Variance")
icc(datalmer1)
performance::icc(datalmer1)
2.1817e-05/(2.1817e-05+2.5471e-02)
```

Plotting graph before fitting

```
datalmerdiag <- data.frame(Residuals=resid(datalmer1),
key=newdata$key,
Fitted= fitted(datalmer1))
ggplot(data=datalmerdiag, aes(x=Fitted,y=Residuals,col=key)) +
geom_point() +
facet_wrap(~key) +
ggtitle("Lowest level residuals facetting by music")

plot_model(datalmer1, type = "re")
```

Fitting

```
datalmer2 <- lmer(danceability~liveness + (1+ liveness|key) ,data=newdata_2)
summary(datalmer2)

datalmer2 <- lmer(danceability~liveness + (1+ liveness|key) ,data=newdata_2,
control = lmerControl(optimizer ="Nelder_Mead"))
summary(datalmer2)
```

Plotting

```
newdata_2$datalmer.predictions <- predict(datalmer2)
ggplot(aes(x = liveness, y = datalmer.predictions,
color = key), data = newdata_2) +
geom_smooth(method = "lm", color ="maroon")+
```

```
geom_point(aes(y = danceability)) +
xlab("Liveness of a song") +
ylab("Danceability of a song") +
ggthemes::theme_tufte()
```

Plotting the lowest residuals

```
datalmer2diag <- data.frame(Residuals=resid(datalmer2),
key=newdata_2$key,
Fitted= fitted(datalmer2))
ggplot(data=datalmer2diag, aes(x=Fitted,y=Residuals,col=key)) +
geom_point() +
facet_wrap(~key) +
ggtitle("Lowest level residuals by song")
```

Plotting Random Effects

```
plot_model(datalmer2, type = "re")
```

Prediction

```
plot_model(datalmer2, type = "eff", terms = "liveness",color = "indianred3") +
geom_point(aes(x=liveness,y=danceability,color="songs"),data=newdata_2)
```

hypothesis

```
library(broom)
shapiro.test(newdata_2$danceability)
shapiro.test(newdata_2$liveness)
#the attributes are not normally distributed
pearson <- cor.test(newdata_2$danceability,newdata_2$liveness,method = "pearson")

tab_pearson <- map_df(list(pearson),tidy)


tab_pearson$estimate<-round(tab_pearson$estimate,digits=2)
tab_pearson$statistic<-round(tab_pearson$statistic,digits = 2)
tab_pearson$p.value<-round(tab_pearson$p.value,digits = 2)
tab_pearson$conf.low<-round(tab_pearson$conf.low,digits = 2)
tab_pearson$conf.high<-round(tab_pearson$conf.high,digits = 2)
tab_pearson$parameter <- NULL


knitr::kable(tab_pearson, caption = "Pearson's Table")
```